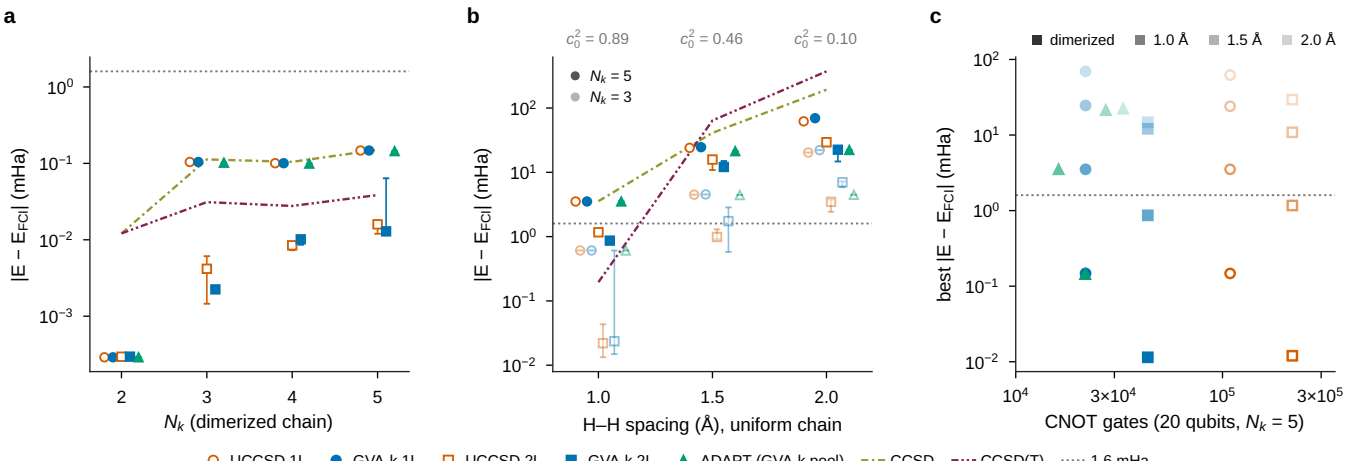
a
b
c
$|E - E_{FCI}|$ (mHa)
N_k (dimerized chain)
$c_0^2 = 0.89$
$c_0^2 = 0.46$
$c_0^2 = 0.10$
$N_k = 5$
$N_k = 3$
H–H spacing (Å), uniform chain
dimerized
1.0 Å
1.5 Å
2.0 Å
best $|E - E_{FCI}|$ (mHa)
CNOT gates (20 qubits, $N_k = 5$)
UCCSD 1L
GVA-k 1L
UCCSD 2L
GVA-k 2L
ADAPT (GVA-k pool)
CCSD
CCSD(T)
1.6 mHa